CLASS101 - TMA00

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Question 1

(a)

Problem

Solve for x:

$$3x + 4 = 10$$

$$\begin{aligned} 3x + 4 &= 10 \\ 3x &= 6 \\ x &= 2 \end{aligned}$$

Question 2

(b)

Problem: Chain Rule in Integration

Evaluate the integral using the chain rule (i.e., u-substitution):

$$\int (3x^2 - 4x)\, e^{\,x^3 - 2x^2}\, \mathrm{d}x$$

We look for an inner function whose derivative appears as a factor.

$$u = x^3 - 2x^2$$
$$\mathrm{d}u = (3x^2 - 4x)\,\mathrm{d}x$$

The differential $\mathrm{d}u$ exactly matches the factor in front of e^u, so the integral collapses to:

$$\begin{aligned}\int (3x^2 - 4x)\, e^{\,x^3 - 2x^2}\,\mathrm{d}x &= \int e^u\,\mathrm{d}u \\ &= e^u + C \\ &= \boxed{e^{\,x^3 - 2x^2} + C}\end{aligned}$$